

INFECTION IN DISGUISE: MULTIMODALITY IMAGING OF A NIPPLE ABSCESS MIMICKING BREAST CANCER

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INTRODUCTION

Non-puerperal nipple and retro-areolar abscesses are uncommon inflammatory breast conditions that may clinically and radiologically mimic malignancy. Suspicious mass-like morphology and enhancement may warrant biopsy to exclude breast cancer.

CASE REPORT

We present a case of 55-year-old woman with a 3-week history of a painful right retro-areolar lump. No fever or preceding trauma. Urgently referred via Pink Ribbon Clinic (PRC) for assessment of suspected malignancy.

Triple assessment was performed.

1. Clinical assessment.

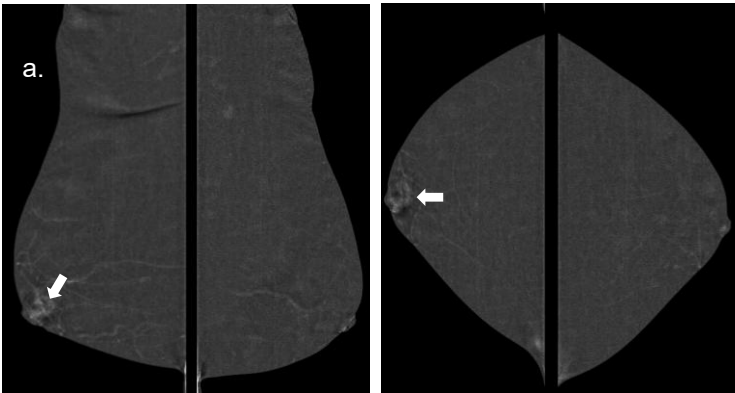
2a. Contrast Enhanced Mammography (CEM).

2b. Targeted Ultrasound Breast.

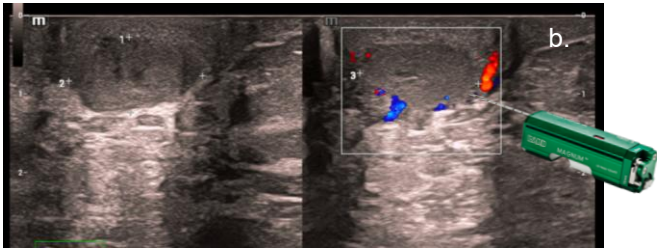
2c. Ultrasound Guided Core needle Breast Biopsy.

3. Histopathology (HPE) results.

IMAGINGS

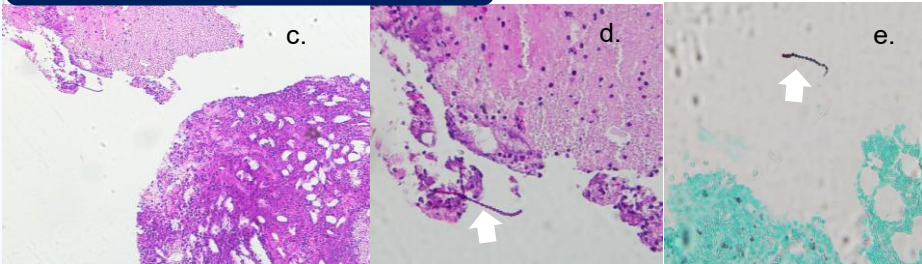


a. Contrast Enhanced Mammography (CEM) showed heterogeneously enhancing focal mass at the right retro-areolar region (shown with white arrow).



b. Targeted ultrasound at the right retro-areolar region, demonstrated oval, circumscribed hypoechoic lesion with peripheral vascularity. This corresponded to the CEM findings. This lesion was biopsied under ultrasound guidance.

HISTOPATHOLOGY

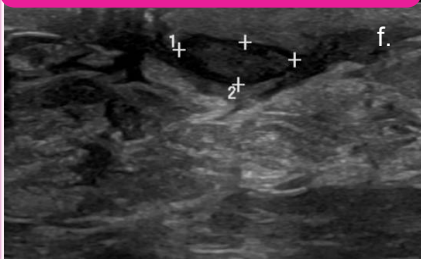


c. At 4x magnification : Periodic acid–Schiff stain (PAS) highlights budding fungal spores within fibrin and neutrophilic exudates. The adjacent fibrofatty tissue is densely infiltrated by neutrophils, forming microabscesses.

d. At 10x magnification: Periodic acid–Schiff stain (PAS) highlights budding yeast cells.

e. At 10x magnification : Grocott's Methenamine Silver (GMS) stain, tissue nearly exhausted but still able to highlight the fungal yeast.

FOLLOW UP IMAGING



f. Following a one-week course of oral antibiotics, repeat ultrasound showed interval reduction in lesion size with partial liquefaction, and the patient reported clinical improvement.

DISCUSSION

This case illustrates that a non-puerperal retro-areolar abscess can closely mimic breast carcinoma on multimodality imaging. Histopathological examination was essential for excluding malignancy and establishing diagnosis. The presence of budding yeast on PAS and GMS stains highlights an uncommon infective finding and emphasizes the importance of clinicopathological correlation.

CONCLUSION

Awareness of this rare radiological mimic and the integration of clinical, imaging, microbiological, and pathological findings are essential for achieving an accurate diagnosis, guiding appropriate management, avoiding unnecessary treatment, and ensuring timely exclusion of underlying breast cancer.

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